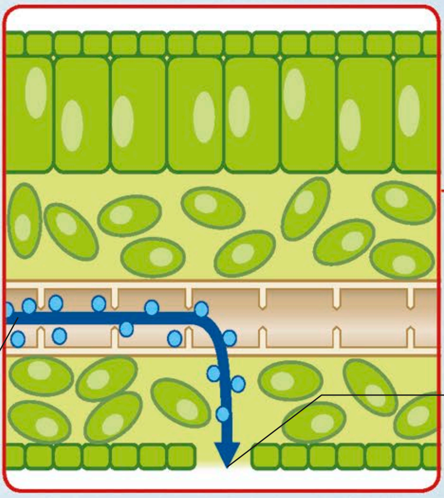
Plant transpiration

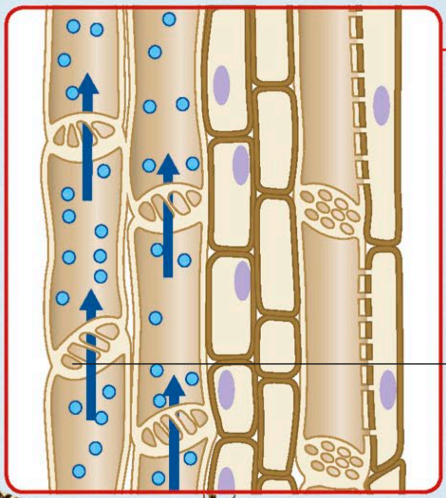
To lift water to their topmost branches, trees need incredible water carrying systems. The tallest ones can pull water with the force of a high-pressure hose.

Plants owe this remarkable ability to impressive engineering. Their trunks and stems are packed with bundles of microscopic pipes. Water and minerals are moved from the soil to the leaves, while food made in the leaves is sent around the entire plant.

3 Pull from above
Water evaporates from the moist tissues inside living leaves. The vapor it generates spills out into the surrounding atmosphere through pores called stomata. This water loss, known as transpiration, is replaced with water arriving from the ground in pipelike xylem vessels.



2 Rising water
The microscopic xylem vessels carry unbroken columns of water through the stem all the way up to the leaves. Water molecules stick together, so as transpiration pulls water into the leaves, all the columns of water rise up through the stem—like water climbing through drinking straws. This is called the transpiration stream.



Transpiration

A tree's water carrying system is incredibly efficient and, unlike similar systems in animals, does not require any energy from the organism. The sun's heat causes water to evaporate from the leaves, a process called transpiration, which triggers the tree to pull more water up from the ground.

Bark
Tough outer layers of bark serve to protect the tree's trunk from injury.

1 Absorption from below
Water seeps into the roots from the soil by a process called osmosis. It then passes into tubes called xylem vessels to join the transpiration stream upward. Microscopic extensions to the root, called root hairs, help maximize the absorption area so the tree can pick up large amounts of water and minerals.